

Exploratory Data Analysis (EDA) Summary Report

Geldium Finance — Credit Card Delinquency Dataset

Prepared as part of the Tata iQ GenAI Powered Data Analytics Forage virtual experience (simulated client engagement).

1. Introduction

This report presents an exploratory data analysis (EDA) of Geldium Finance's customer dataset, conducted to assess data quality and surface patterns relevant to credit card delinquency risk ahead of predictive modeling. The analysis was performed directly on the provided dataset of 500 customer records using Python (pandas), with structuring and narrative support from generative AI tools.

The goal of this analysis is threefold: (1) confirm the dataset is fit for modeling, (2) identify and document data quality issues that must be resolved before modeling, and (3) surface any early risk signals — while being transparent about where the data does, and does not, support strong conclusions.

2. Dataset Overview

The dataset contains 500 customer records across 19 fields, combining demographic, financial, and behavioral attributes with a 6-month payment history and a binary delinquency label.

Attribute	Detail
Number of records	500
Number of fields	19
Target variable	Delinquent_Account (0 = No, 1 = Yes)
Overall delinquency rate	16.0% (80 of 500 customers)
Numerical features	Age, Income, Credit_Score, Credit_Utilization, Missed_Payments, Loan_Balance, Debt_to_Income_Ratio, Account_Tenure
Categorical features	Employment_Status, Credit_Card_Type, Location, Month_1–Month_6 (On-time / Late / Missed)
Duplicate records	0 duplicate rows; 0 duplicate Customer_IDs

Feature ranges (sanity check against expected bounds)

Field	Observed Range	Notes
Age	18 – 74	Within plausible bounds
Credit_Score	301 – 847	Within the standard 300–850 scale
Credit_Utilization	0.05 – 1.026	4 records exceed 1.0 (100%) — see Section 4
Debt_to_Income_Ratio	0.10 – 0.55	Within plausible bounds
Account_Tenure	0 – 19 years	Within plausible bounds

3. Missing Data Analysis

Three fields contain missing values. None of the categorical or target fields have missing entries.

Field	Missing Count	% of Records	Recommended Treatment
Income	39	7.8%	Median imputation (numeric, moderately right-skewed)
Loan_Balance	29	5.8%	Median imputation; flag for review given pattern below
Credit_Score	2	0.4%	Median imputation (negligible volume)

Per the Imputation Guide, median was chosen over mean for all three fields because each shows meaningful spread and the potential for skew (e.g., Income ranges from ~\$15K to ~\$200K); median is more robust to that spread than a mean would be.

Missingness pattern check

Missingness was cross-checked against the delinquency label to rule out an obvious Missing Not at Random (MNAR) pattern:

- Income missing: delinquency rate is 12.8% among those records vs. 16.3% overall — no evidence income-missingness concentrates in delinquent accounts.
- Loan_Balance missing: delinquency rate is 24.1% among those records vs. 15.5% for complete records — a notable gap, though based on only 29 records, so it should be treated as a lead to monitor after imputation rather than a confirmed pattern.

4. Key Findings and Risk Indicators

4.1 Data quality issues identified

- Employment_Status is inconsistently encoded: the same category appears as “Employed”, “employed”, and “EMP” (240 records once normalized), alongside “Unemployed” (93), “retired” (87), and “Self-employed” (80). This must be standardized before it is used as a model feature or segmentation variable.
- 4 records (0.8%) show Credit_Utilization above 100% (up to 102.6%), which is out of range for a utilization ratio and should be capped, flagged, or investigated as a data entry issue rather than dropped outright.
- No duplicate customers and no out-of-range values were found for Age, Credit_Score, or Debt_to_Income_Ratio.

4.2 Correlation with delinquency

Pearson correlation was computed between each numeric feature and Delinquent_Account. Contrary to the assumption that any single factor would dominate, no individual numeric feature shows a strong linear relationship with delinquency in this dataset — all correlations fall below 0.05 in magnitude:

Feature	Correlation with Delinquent_Account
Income	0.045
Credit_Score	0.035
Credit_Utilization	0.034
Debt_to_Income_Ratio	0.034
Age	0.023
Account_Tenure	−0.040
Missed_Payments	−0.026
Loan_Balance	−0.004

This is a genuinely important finding for modeling: it suggests delinquency in this dataset is not well explained by any single linear driver, and a model will likely need to rely on combinations of features (interactions) rather than one dominant signal. It also means claims of a “strong correlation” between any one variable and delinquency are not supported by this data and should not be asserted in downstream reporting.

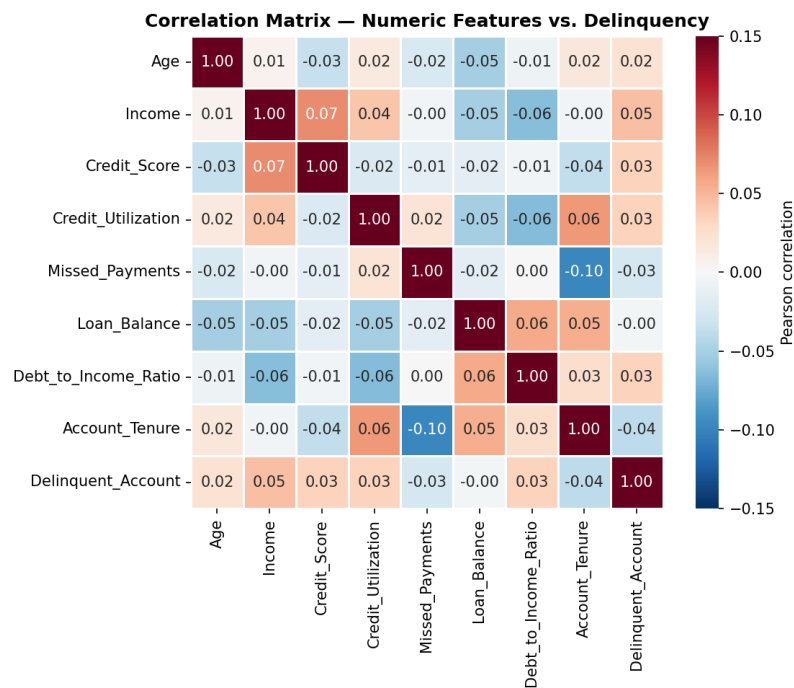


Figure 1. Correlation matrix confirming no numeric feature exceeds 0.05 in magnitude with Delinquent_Account.

4.3 Delinquency rate by segment (directional, not strongly predictive on their own)

Grouping customers into buckets shows mild directional patterns worth carrying into modeling, even though none rise to a strong standalone signal:

Segment	Delinquency Rate	Sample Size
Debt-to-Income 35–50%	18.5%	135
Debt-to-Income 20–35%	17.0%	276
Debt-to-Income under 20%	9.8%	82
Credit Utilization 30–50%	17.0%	182
Credit Utilization under 30%	12.0%	83
Age 46–60	21.7%	138
Age 31–45	12.3%	138
Unemployed (normalized)	19.4%	93
Retired (normalized)	11.5%	87

Higher debt-to-income ratio shows the clearest directional trend (9.8% → 18.5% delinquency as DTI rises), followed by employment status (unemployed customers delinquent at nearly double the rate of retirees). Age shows a non-linear pattern, with the 46–60 group elevated relative to neighboring age bands. Missed_Payments and months-missed-in-history did not show a clear monotonic relationship with the label in this sample.

4.4 Combined Risk Segment Analysis (Employment × Debt-to-Income Ratio)

Since no single numeric feature showed a strong linear relationship with delinquency (Section 4.2), employment status was analyzed jointly with debt-to-income ratio to test for an interaction effect.

Employment Status	DTI < 35%	DTI ≥ 35%
Employed	16.4% (n=165)	16.0% (n=75)
Self-employed	13.1% (n=61)	26.3% (n=19)
Retired	14.5% (n=62)	4.0% (n=25)
Unemployed	15.7% (n=70)	30.4% (n=23)

The combined segment of unemployed customers with DTI ≥ 35% shows the highest observed delinquency rate in the dataset (30.4%, n=23) — nearly double the portfolio average of 16.0%. This finding informed the targeted pilot recommendation in Task 3.

Within this sample, the interaction is not uniform across employment categories. High DTI raises delinquency risk for Unemployed and Self-employed customers (to 30.4% and 26.3% respectively) but has little effect on Employed customers (16.0%, essentially flat vs. their low-DTI peers at 16.4%) and appears to lower observed risk among Retired customers with high DTI (4.0%, n=25) — though this last cell is small and should be treated as directional rather than conclusive, consistent with the same caution applied to the Loan_Balance missingness pattern in Section 3. This suggests the risk signal is concentrated in employment categories tied to income instability, not debt burden alone.

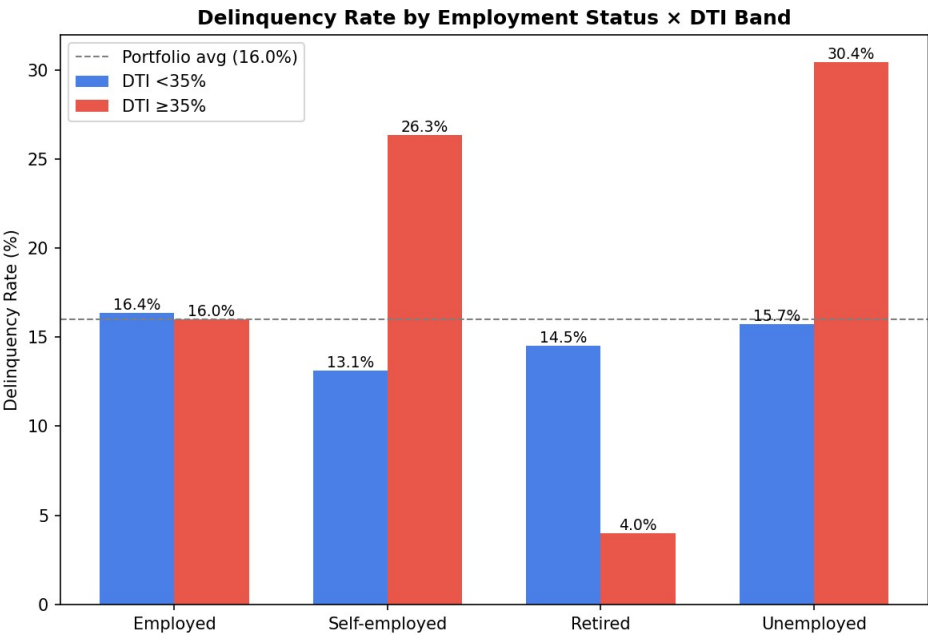


Figure 2. Delinquency rate by Employment Status × DTI band, visualizing the Section 4.4 interaction analysis.

5. AI & GenAI Usage

Generative AI (Claude) was used to help structure this report, draft narrative explanations of the statistical output, and cross-check the interpretation of correlation and bucket results against standard credit-risk EDA practice. All figures in this report were computed directly from the dataset in Python (pandas); AI was not used to generate or estimate any numeric values.

Example prompts used

- “Run descriptive statistics and missing-value counts on this dataset and summarize data quality issues.”
- “Compute correlation between each numeric feature and the delinquency label and explain what the results imply for modeling.”
- “Check whether missingness in Income and Loan_Balance is related to the delinquency outcome.”

6. Conclusion & Next Steps

The dataset is broadly complete and usable for modeling once two data quality issues are resolved: standardizing Employment_Status categories and addressing the 4 out-of-range Credit_Utilization values. Missing values in Income, Loan_Balance, and Credit_Score are limited in volume (under 8% each) and suitable for median imputation.

The most important analytical finding is that no single numeric feature strongly predicts delinquency in isolation — debt-to-income ratio and employment status show the clearest directional patterns, but a predictive model will need to combine multiple features rather than rely on one dominant driver.

Recommended next steps

- Standardize Employment_Status categories (Employed / Unemployed / Self-employed / Retired) before feature engineering.
- Cap or flag Credit_Utilization values above 1.0 rather than treating them as valid inputs.
- Apply median imputation to Income, Loan_Balance, and Credit_Score; monitor the Loan_Balance missingness pattern after imputation.
- Proceed to predictive modeling (Task 2) using Debt-to-Income Ratio, Credit Utilization, Employment_Status, Age, and Missed_Payments as candidate features, with attention to feature interactions rather than single-variable effects.